

Oil Pan

 [Ford Ecat Electronic Parts Catalog](#)

Material	
Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS- M2C945-A
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4
Motorcraft® Silicone Gasket Remover ZC-30-A	—

Removal

All vehicles

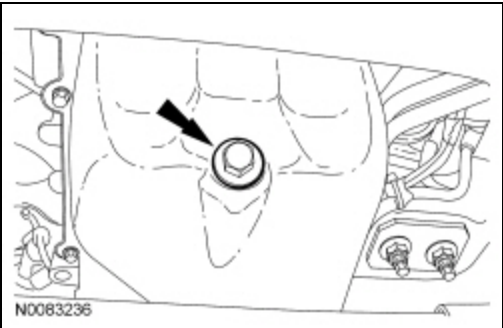
1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.

Four-Wheel Drive (4WD) vehicles

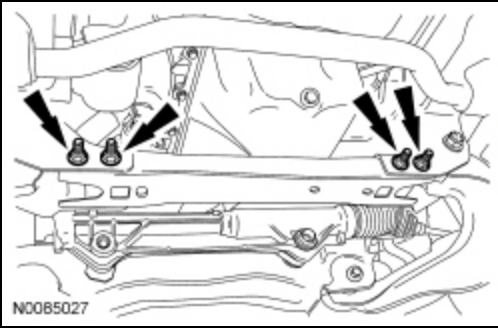
2. Remove the 4 bolts and the skidplate.

All vehicles

3. Remove the oil drain plug and drain the engine oil. Install the drain plug when finished.
 - Tighten to 23 Nm (17 lb-ft).

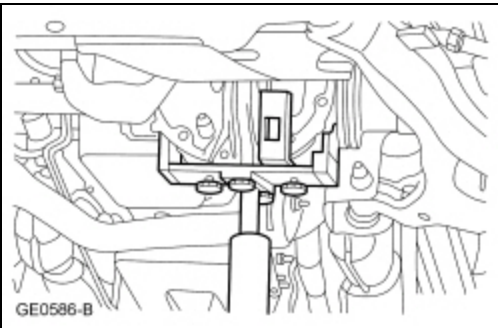


4. Remove the 4 nuts, the 4 bolts and the crossmember.



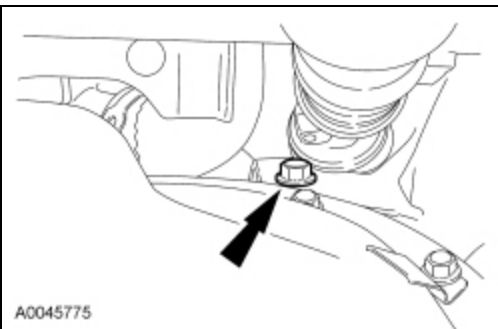
Four-Wheel Drive (4WD) vehicles

5. Position a suitable hydraulic jack under the front axle. Securely strap the jack to the axle.

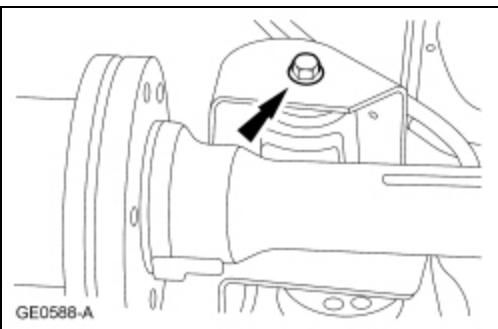


6. **NOTE:** Rotate the steering column so the pinch bolt for the steering column coupling allows clearance for the isolator bolt.

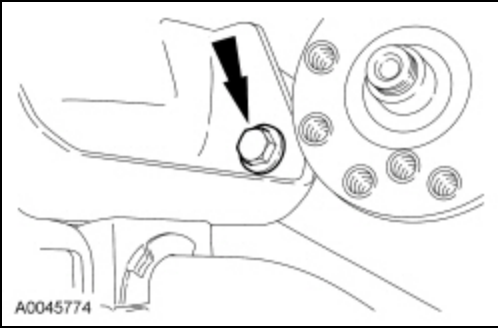
Remove the upper front axle carrier mounting bushing bolt.



7. Remove the axle shaft housing carrier bushing bolt.



8. Remove the lower front axle carrier mounting bushing bolt.

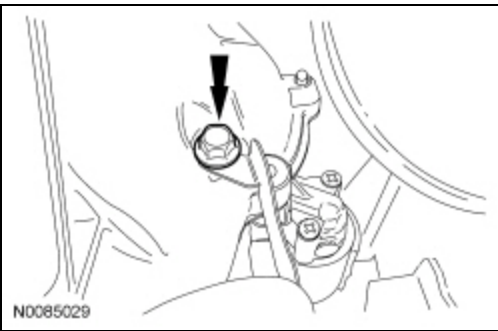


9. **NOTICE:** Use care when lowering the front axle housing or the vacuum lines to the axle solenoid may become disconnected or damaged.

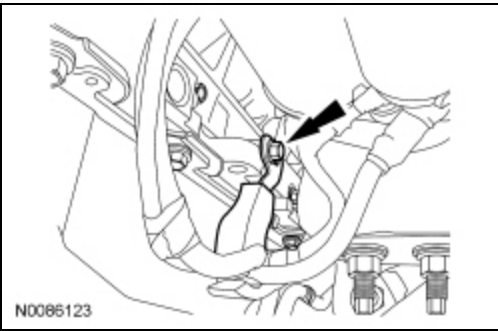
Lower the axle to allow clearance for the oil pan to be removed.

All vehicles

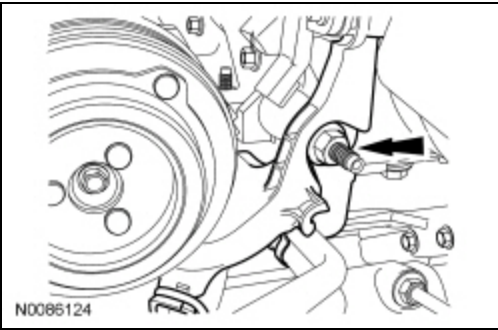
10. Remove the bolt and position the transmission fluid cooling tubes rear support bracket aside.



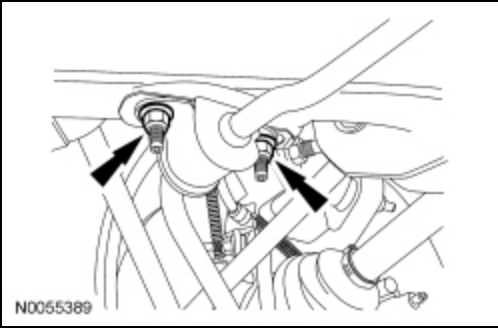
11. Remove the starter wiring harness rear support bracket pushpin.



12. Remove the nut and position the starter wiring harness and the transmission fluid cooler tubes aside.



13. Remove and discard the 4 stabilizer bar bracket nuts and allow the stabilizer bar to swing downward.



14. Remove the 20 bolts, the oil pan and the gasket.

- Discard the gasket.

15. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

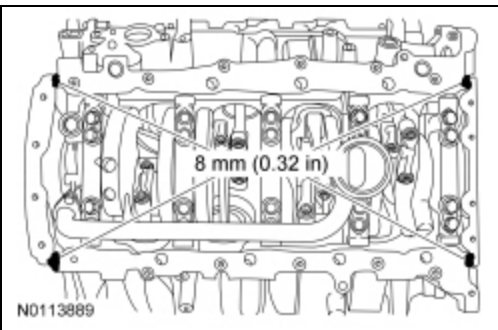
Inspect the oil pan. Clean the gasket mating surfaces of the oil pan and engine block with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

Installation

All vehicles

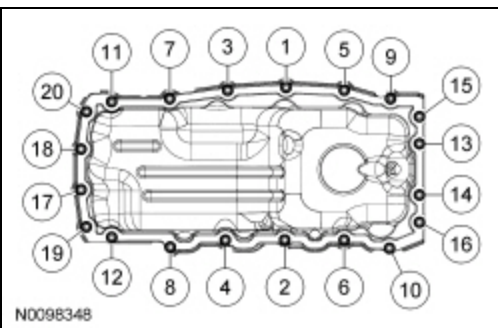
1. **NOTE: If not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.**

Apply silicone gasket and sealant at the crankshaft rear seal retainer plate-to-cylinder block sealing surface and at the engine front cover-to-cylinder block sealing surface.



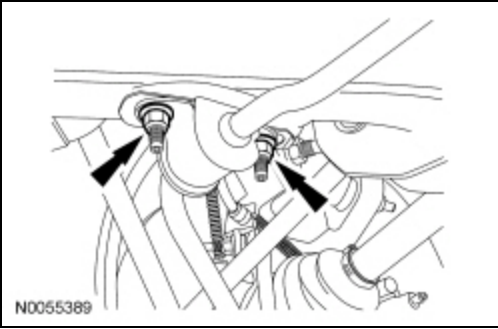
2. Position a new gasket and the oil pan and install the 20 bolts.

- Tighten the bolts in the sequence shown in 3 stages.
 - Stage 1: Tighten to 2 Nm (18 lb-in).
 - Stage 2: Tighten to 10 Nm (89 lb-in).
 - Stage 3: Tighten an additional 45 degrees.

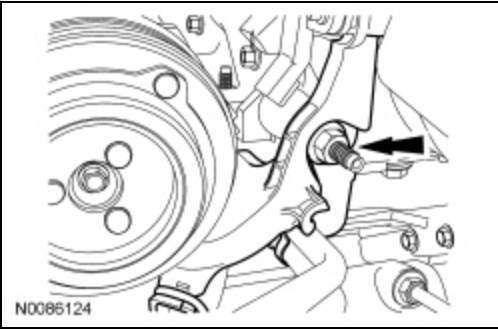


3. Position the swaybar and install the 4 new stabilizer bar brackets nuts.

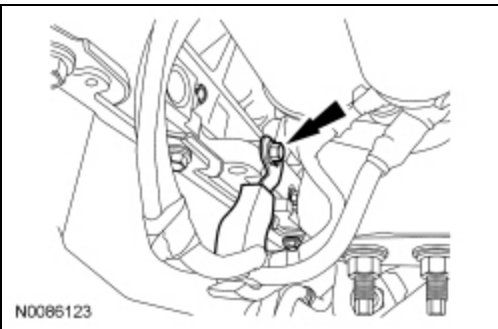
- Tighten to 55 Nm (41 lb-ft).



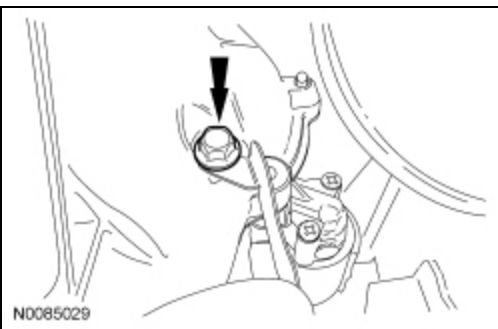
4. Position the transmission fluid cooler tube support bracket, the starter wiring harness support bracket and install the nut.
 - Tighten to 10 Nm (89 lb-in).



5. Position the starter wiring harness and install the starter wiring harness rear support bracket pushpin.



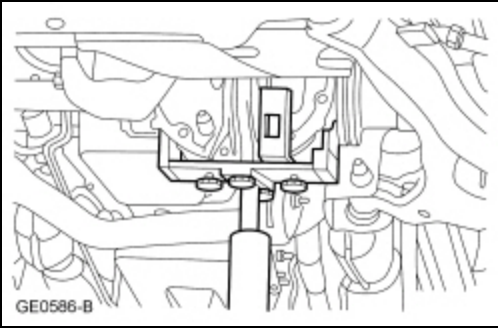
6. Install the transmission fluid cooling tubes rear support bracket bolt.
 - Tighten to 48 Nm (35 lb-ft).



Four-Wheel Drive (4WD) vehicles

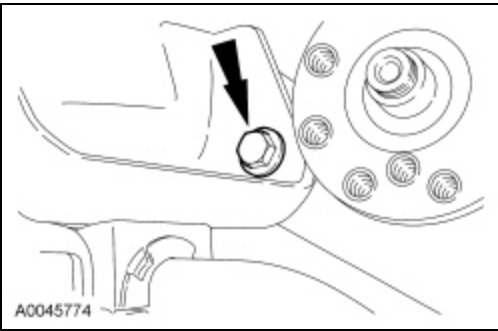
7. **NOTICE:** Use care when positioning the front axle housing or the vacuum lines to the axle solenoid may become disconnected or damaged.

Raise the front axle carrier into position.



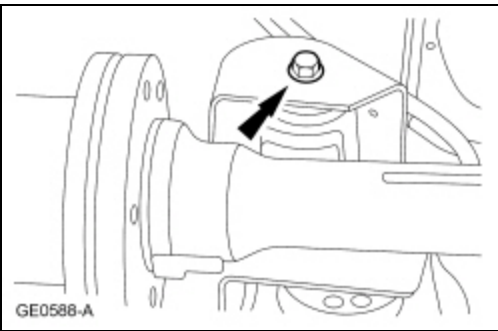
8. Install the lower front axle carrier mounting bushing bolt.

- Tighten to 115 Nm (85 lb-ft).



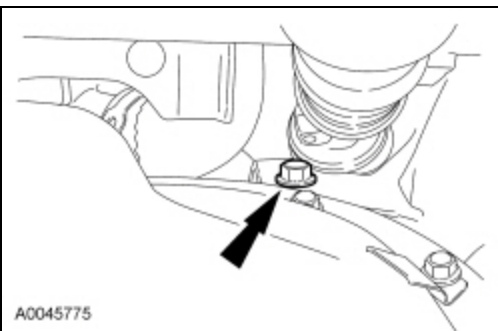
9. Install the axle shaft housing carrier bushing bolt.

- Tighten to 115 Nm (85 lb-ft).



10. Install the upper front axle carrier mounting bushing bolt.

- Tighten to 115 Nm (85 lb-ft).



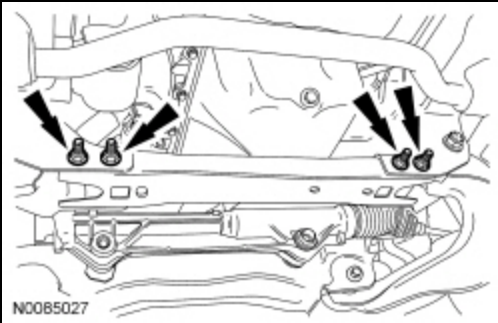
11. Install the skid plate and 4 bolts.

- Tighten to 48 Nm (35 lb-ft).

All vehicles

12. Position the crossmember and Install the 4 bolts and the 4 nuts.

- Tighten to 90 Nm (66 lb-ft).



13. Fill the engine with clean engine oil.